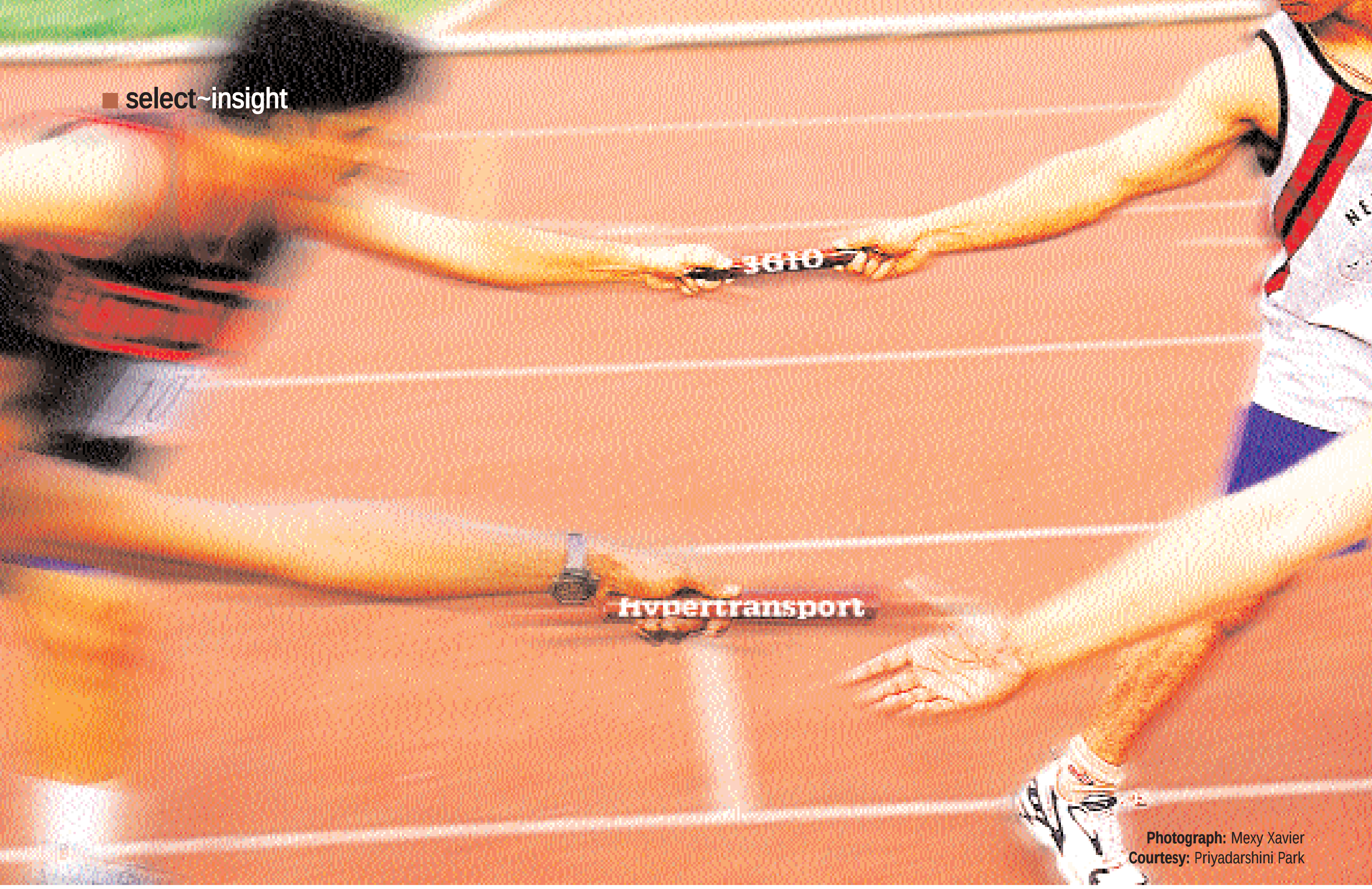

Photograph: Mexy Xavier
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HyperTransport vs 3GIO

As the battle for CPU supremacy wages on, Intel and AMD fight to promote their proposals to rework the innards of computers

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There is no doubt about it. The escalation seen in today's CPU speeds is stunning. Hard disks are buzzing along faster than ever and the 3D technology that's available today was dismissed as fantasy a decade ago. However, without due consideration, pushing barriers in clock speeds, storage densities and fill-rates will be in vain. Tomorrow's devices are going to need infrastructure suited for their requirements. Without high-bandwidth protocols in place, the new technology is virtually useless.

Long-time rivals, Intel and AMD have each introduced blueprints of a new I/O bus—a successor to the aging PC architecture, the PCI bus—aimed at handling future bandwidth requirements and which are scalable at the same time. AMD's HyperTransport and Intel-backed 3GIO (known as Arapahoe) each proposes to change one of the most overlooked parts in a PC, the I/O bus.

But before we discuss the technologies involved in HyperTransport and 3GIO, let's rewind a little and see the limitations in the existing technology.

Beyond the confines of the PC

The components of a PC communicate by sending and receiving data through an interconnected network of wires called buses. A bus is essentially a high-speed conduit for digital information to travel. Several types of buses exist and each operate at a different speed and protocol, depending upon the standard at which the bus functions.

The PCI bus is responsible for transporting